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Mission Requirements Document

ECSS-E-ST-10-06C

SpaceCDF

2026

Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

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Document Information

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Change Record

Issue	Date	By	Summary of changes
1.0	2026-05-11	SpaceCDF	Initial issue of Mission Requirements Document.

Acknowledgement — Generative AI (AIG)

This document was produced with the assistance of generative AI as part of the SpaceCDF Concurrent Design Facility workflow. The design-loop convergence, agent rationales, embedded figures and document rendering are generated by the SpaceCDF backend (Python · matplotlib · python-docx) guided by ECSS, NASA SEH and SMAD-4 references. Editorial framing, worked examples and pedagogical commentary remain owned by the SpaceCDF teaching team.

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Acronyms & Abbreviations

Acronym	Definition
MRD	Mission Requirements Document

Applicable Documents

ID	Reference	Title
AD-01	ECSS-M-ST-10C Rev.1	Space project management

Reference Documents

ID	Reference	Title
RD-01	SMAD-4	Space Mission Analysis and Design

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1 Mission Need

EOSAT-1 addresses **persistent demand** for high-frequency ocean colour observation.

2 Mission Objectives

Objective	Type	Target
4-band ocean colour	MoP	<2 day revisit